

SSC CGL TIER-1: SPEED & TRAP ELIMINATION BIBLE

THE 1–10 SECOND SPEED WEAPONS & REVERSE-ENGINEERED TRAPS PLAYBOOK

Exhaustive Question Taxonomy Across All 4 Sections • Concrete Live Examples • 1-Sentence Memory Acronyms • Speed Tiers
(1–2s, 2–5s, 5–10s)

1. THE EXAMINER'S TRAP ARCHITECTURE (HOW PAPERS ARE MADE)

Examiners under TCS use automated question-bank generators with human distractor-tuning. Every distractor option is deliberately designed to capture a specific cognitive failure under the 36-second time pressure:

Trap Mechanism	How the Examiner Builds It	The 1–2 Second Counter-Attack
The Camouflage Twins	Calculates correct answer (+144 or 5/12), then adds (-144 or 12/5) to catch sign/inversion slips.	If options contain exact inverse/opposite pairs, eliminate the other two options immediately .
The Halfway Pitstop	Solves 50% of the problem (e.g. finds total work instead of remaining work) and puts it as Option A .	Always read the last 3 words of the question before clicking Option A.
The Extreme Outlier	Inserts one abnormally high or low number (e.g. 24, 26, 28, 148) to catch desperate blind guessers.	Outliers are 95% false. Cross out the extreme and choose between clustered medians.
The Pi (π) Multiplier	Mensuration formulas contain π ($22/7 = 2 \times 11$). Wrong options are made without 11 divisibility.	11-Divisibility Filter: (Odd Sum - Even Sum) must equal 0 or 11. 3 options fail in 2 seconds.
The Preposition Cloak	Hides root singular subjects behind long plural prepositional phrases (e.g., 'The quality of apples were...').	Blindfold words inside 'of / with / along with' and connect root subject directly to verb.
The Ego-Burner Puzzle	Places 1 complex non-linear number series in Reasoning to trap high-scorers into wasting 3 minutes.	20-Second Rule: If logic doesn't jump out in 20s, flag and skip immediately.

2. REASONING: CUBE & DICE MASTERPLAY (ALL 5 TYPES)

Type 1: Two Dice Views with ONE Common Face	Speed Target: 2-3s
<i>Memory Acronym: "Lock the pivot, spin like a clock."</i>	
Examiner's Trap: Rotates remaining numbers into non-standard orientations to force 3D visualization.	
Speed Weapon: Write clockwise sequence from common number for both dice. Aligned columns are opposite pairs.	
Live Example & Execution: Dice 1 shows (3, 5, 1). Dice 2 shows (3, 2, 6). Write: 3 → 5 → 1 and 3 → 2 → 6. Result: 5 is opposite 2, 1 is opposite 6, 3 is opposite 4. (Solved in 2s)	
Type 2: Two Dice Views with TWO Common Faces	Speed Target: 1-2s
<i>Memory Acronym: "Kill the twins, the lonely wins."</i>	
Examiner's Trap: Shows two views where two faces match (e.g. 2 and 5), baiting you to draw clockwise circles.	
Speed Weapon: Direct Cancellation: Cancel out the two common faces. The remaining distinct face on Dice 1 is ALWAYS opposite the remaining face on Dice 2.	
Live Example & Execution: Dice 1 shows (2, 5, 3). Dice 2 shows (2, 5, 6). Cancel 2 and 5 on both → 3 is opposite 6 instantly. Zero pen needed! (Solved in 1s)	
Type 3: Unfolded / Open Flat Cube	Speed Target: 1-2s
<i>Memory Acronym: "Jump one block in a line, opposite pairs you will find."</i>	
Examiner's Trap: Folds flat boxes into multiple 3D option cubes, hoping you fail to track adjacent vs opposite surfaces.	
Speed Weapon: Alternate-Box Rule: Every 2nd box in a straight line is opposite. Opposite faces can NEVER appear together on a visible folded cube.	

Live Example & Execution: Unfolded strip shows 1-2-3 in a line. 1 is opposite 3. Eliminate any option cube showing both 1 and 3 together.
(Solved in 2s)

Type 4: Painted Cube Cut into Smaller Cubes ($n \times n \times n$)

Speed Target: 3-5s

Memory Acronym: "Corners are 8, Edges $12(n-2)$, Faces $6(n-2)^2$, Core $(n-2)^3$."

Examiner's Trap: Asks for count of 0, 1, 2, or 3-face painted cubes when a large painted cube is cut into N pieces.

Speed Weapon: Find $n = \sqrt[3]{\text{Total. 3-Painted} = 8 \text{ (always); 2-Painted} = 12(n-2); 1\text{-Painted} = 6(n-2)^2; 0\text{-Painted} = (n-2)^3}$.

Live Example & Execution: 125 smaller cubes $\rightarrow n=5$. (1) 0-painted = $(5-2)^3 = 27$. (2) 1-painted = $6(5-2)^2 = 54$. (3) 2-painted = $12(5-2) = 36$.
(Solved in 4s)

Type 5: Multi-Colored Painted Cube (Red, Blue, Green on Pairs)

Speed Target: 5-8s

Memory Acronym: "Target edges/faces only, ignore full cube formulas."

Examiner's Trap: Opposite pairs painted with different colors (Red, Blue, Green). Asks: 'How many cubes have ONLY Red & Blue?'

Speed Weapon: Red and Blue meet at exactly 4 edges (out of 12). Formula: $4 \times (n - 2)$.

Live Example & Execution: Cube cut into 64 pieces ($n=4$). Cubes with Red & Blue only = $4 \times (4 - 2) = 8$. (Solved in 5s)

3. QUANTITATIVE APTITUDE: THE 10-SECOND SPEED ARSENAL

Q-1: Mensuration Area & Volume (π Filter)

Speed Target: 1-2s

Memory Acronym: "When Pi is inside, Eleven will decide."

Examiner's Trap: Gives composite figures (cylinder + cone) with ugly decimal radii (e.g. $r=10.5$, $h=14$) to force 3-minute long calculations.

Speed Weapon: Formula contains π ($22/7 = 2 \times 11$). Test options for divisibility by 11: (Odd Sum - Even Sum) = 0 or 11.

Live Example & Execution: Options: A) 736, B) 748, C) 752, D) 745.

Test B: $748 \rightarrow (7+8) - 4 = 11$. Divisible by 11! Option B is correct without writing a formula! (Solved in 2s)

Q-2: Percentage / Price & Consumption (AB Ladder)

Speed Target: 1-2s

Memory Acronym: "Up by $+a/b$, down by $-a/(a+b)$."

Examiner's Trap: Price of commodity increases by x%, finds required consumption drop to keep expense constant.

Speed Weapon: Convert % to fraction. If increase is $+a/b$, required reduction is ALWAYS $-a/(a+b)$. No variables needed!

Live Example & Execution: Price rises by 25% ($+1/4$) $\rightarrow$ Reduction = $1/(4+1) = 1/5 = 20\%$. If price falls by 20% ($-1/5$) $\rightarrow$ Consumption rises by $1/(5-1) = 1/4 = 25\%$. (Solved in 2s)

Q-3: Profit, Loss & Marked Price (The MP/CP Ratio)

Speed Target: 3-5s

Memory Acronym: "MP over CP equals $(100+P\%)$ over $(100-D\%)$."

Examiner's Trap: Gives discount% and profit% together, asks for MP when CP is given or vice versa.

Speed Weapon: Direct Proportionality: $MP / CP = (100 + P\%) / (100 - D\%)$.

Live Example & Execution: Dealer gives 10% discount and makes 20% profit. CP = 300.

$MP/CP = (100+20)/(100-10) = 120/90 = 4/3$. Since CP=300 (3×100), MP = $4 \times 100 = 400$. (Solved in 4s)

Q-4: CI vs SI Difference (2-Year & 3-Year)

Speed Target: 3-5s

Memory Acronym: "2-Yr: $P(R/100)^2$ | 3-Yr: $\text{Diff} \times (300+R)/100$."

Examiner's Trap: Gives CI and SI difference for 2 or 3 years at R%, asks to find Principal P.

Speed Weapon: 2 Years: $\text{Diff} = P(R/100)^2$.

3 Years: $\text{Diff} = P(R/100)^2 \times [(300+R)/100]$.

Live Example & Execution: Diff between CI and SI for 2 yrs at 10% is Rs 100.

$100 = P(10/100)^2 \rightarrow 100 = P(1/100) \rightarrow P = \text{Rs } 10,000$. (Solved in 4s)

Q-5: Time, Speed & Distance: Early vs Late Shortcut

Speed Target: 3-5s

Memory Acronym: "Product over Delta-Speed times Time-Gap."

Examiner's Trap: Moving at S1 reaches late; moving at S2 reaches early. Finds distance.

Speed Weapon: Distance = $[S1 \times S2 / |S1 - S2|] \times \Delta T$ (hours). Add late+early times to get total gap.

Live Example & Execution: At 4 km/h is 10 min late; at 5 km/h is 5 min early. Gap = 15 min (15/60 hr).
 $D = [4 \times 5 / (5-4)] \times (15/60) = 20 \times (1/4) = 5 \text{ km}$. (Solved in 4s)

Q-6: Time & Work: 'Leaving Before Completion' Addition

Speed Target: 5-8s

Memory Acronym: "If he tries to leave before the end, add his work and keep your friend."

Examiner's Trap: A worker leaves 2 or 3 days *before* completion, making reverse-calculation tedious.

Speed Weapon: Do NOT subtract! Add the leaving worker's missed work to Total Work, then divide by (Eff_A + Eff_B).

Live Example & Execution: A takes 10d, B takes 15d. B leaves 2d before completion. LCM=30. Eff(A)=3, Eff(B)=2.
 B's 2 days = 4 units. Total = 30+4 = 34. Days = 34/(3+2) = **6.8 days**. (Solved in 5s)

Q-7: Algebra: Value Putting & Degrees of Freedom

Speed Target: 2-3s

Memory Acronym: "More unknowns than lines? Put zero or one and it shines."

Examiner's Trap: Gives symmetric condition like $a+b+c=0$, asks for massive fraction ($a^2/bc + b^2/ca + c^2/ab$).

Speed Weapon: Substitute simple integers satisfying the equation. Put $a=1$, $b=1$, $c=-2$ (Sum=0).

Live Example & Execution: Expression = $(1/-2) + (1/-2) + (4/1) = -0.5 - 0.5 + 4 = 3$. Solved in 2 seconds!

Q-8: Trigonometry: The 0° / 45° / 90° Angle Substitution

Speed Target: 2-4s

Memory Acronym: "Sin-Cos loves 90, Tan-Cot loves 45."

Examiner's Trap: Complex identity simplification like $(\sin\theta / 1+\cos\theta) + (1+\cos\theta / \sin\theta)$.

Speed Weapon: If only sin/cos appear → plug $\theta = 90^\circ$ ($\sin 90=1$, $\cos 90=0$). If tan/cot appear → plug $\theta = 45^\circ$.

Live Example & Execution: Put $\theta = 90^\circ$: $(1 / 1+0) + (1+0 / 1) = 1 + 1 = 2$. Check option 2 $\csc(90^\circ) = 2(1) = 2$. Option matches! (Solved in 3s)

Q-9: Number System: 72 & 88 Divisibility (Tail-Head Split)

Speed Target: 5-8s

Memory Acronym: "Check the Tail for 8, sum the Head for 9."

Examiner's Trap: 9-digit number 789x5432y is divisible by 72 (8×9). Finds $(2x + 3y)$.

Speed Weapon: (1) Tail: Last 3 digits (32y) must be div by 8 → $y=8$. (2) Head: Sum of digits div by 9 → $46+x=54$ → $x=8$.

Live Example & Execution: Value = $2(8) + 3(8) = 16 + 24 = 40$. (Solved in 6s)

Q-10: Digital Root (Mod-9) & Unit Digit Truncation

Speed Target: 2-4s

Memory Acronym: "Sum the digits to one, throw the nines and you're done."

Examiner's Trap: Huge multiplication/simplification problem (e.g. $412 \times 318 - 183$).

Speed Weapon: Sum all digits of numbers into a single digit ($9=0$). Invariant under +, -, ×.

Live Example & Execution: $412 \times 318 \rightarrow (4+1+2) \times (3+1+8) = 7 \times 12(3) = 21 \rightarrow 3$. Only the option with Digital Root 3 is correct! (Solved in 3s)

4. ENGLISH COMPREHENSION: SPEED TACTICS

E-1: Active & Passive Voice (Tense-Immunity Verb Scan)

Speed Target: 1-2s

Memory Acronym: "Voice changes the costume, NEVER the tense."

Examiner's Trap: Long 3-line sentences with passive distractors altering past tense into present/future.

Speed Weapon: Read ONLY the verb!

- V2 → was/were + V3
- has/have + V3 → has/have been + V3
- is/are + V-ing → is/are being + V3

Live Example & Execution: Sentence: 'The chef prepared a sumptuous dinner.'

Scan: V2 (prepared) → must have 'was prepared'. Pick option with 'was prepared' in 1s!

E-2: Direct & Indirect Speech (Said-Backshift Scan)**Speed Target: 1-2s***Memory Acronym: "Past reporting verb kicks every tense one step back."***Examiner's Trap:** Complex dialogues with multiple pronoun shifts and modal changes.**Speed Weapon:** When reporting verb is 'Said', scan tense backshift: Present → Past; Past Simple → Had + V3; Will → Would; Can → Could.**Live Example & Execution:** Sentence: He said, 'I am watching a movie.'

Scan: 'am watching' → must become 'was watching'. Match in 1 second!

E-3: Para Jumbles / PQRS (Mandatory Noun-Pronoun Hooks)**Speed Target: 3-5s***Memory Acronym: "Find the Name before the He/She/It."***Examiner's Trap:** 4 long historical/scientific paragraphs taking 60 seconds to read fully.**Speed Weapon:** Locate Mandatory Pair: Noun introduces subject (Sentence B: 'Isaac Newton...'), Pronoun follows (Sentence D: 'He formulated...'). Pair = BD. Scan options for 'BD'.**Live Example & Execution:** Options: A) BACD, B) BDA■, C) CBDA, D) ADBC. Only B and C have BD. Test opening sentence → Pick answer in 4s!**E-4: Spotting Errors (Preposition-Fluff Cloak)****Speed Target: 1-3s***Memory Acronym: "Blindfold words between prepositions, match the root subject to the verb."***Examiner's Trap:** Places plural nouns right before verbs to camouflage singular subject agreement (e.g. 'The quality of these apples were poor').**Speed Weapon:** Ignore everything inside prepositional phrases ('of...', 'along with...', 'together with...'). Match root noun to verb directly.**Live Example & Execution:** Read: 'The quality [of these apples] were poor' → 'The quality were' is WRONG → 'The quality was'. (Solved in 2s)**E-5: Conjunction Inversion Traps****Speed Target: 1-2s***Memory Acronym: "Hardly pairs with When; No Sooner takes Than; Lest takes Should."***Examiner's Trap:** Mixes conjunction pairs (Hardly...than, No sooner...when, Lest...may).**Speed Weapon:** Lock fixed pairs: Hardly/Scarcely → WHEN; No Sooner → THAN; Lest → SHOULD.**Live Example & Execution:** Sentence: 'No sooner had he arrived when the bell rang.' → Error! 'when' must be 'than'. (Solved in 1s)**5. GENERAL AWARENESS: 5-SECOND ELIMINATION PROTOCOL****G-1: Delhi Sultanate Chronology (SKTSL Ladder)****Speed Target: 1-2s***Memory Acronym: "Some Kings Talked So Loud."***Examiner's Trap:** Asks for chronological order of rulers, dynasties, or architectural monuments.**Speed Weapon:** SKTSL: Slave (1206) → Khilji (1290) → Tughlaq (1320) → Sayyid (1414) → Lodi (1451).**Live Example & Execution:** Eliminate any monument built by Tughlaq showing up before Khilji (e.g. Tughlaqabad before Alai Darwaza). (Solved in 2s)**G-2: Bio-Medical & Science Prefixes****Speed Target: 1-2s***Memory Acronym: "The prefix tells the organ, don't read the whole disease."***Examiner's Trap:** Gives complex medical terms and asks which organ is affected.**Speed Weapon:** • Hepato- = Liver | • Nephro- = Kidney | • Osteo- = Bone | • Rhino- = Nose | • Myo- = Muscle | • Derm- = Skin.**Live Example & Execution:** Hepatitis → Liver; Nephritis → Kidney; Osteomalacia → Bones. Match directly! (Solved in 1s)**G-3: Classical Dance & Gharana Keywords****Speed Target: 1-2s***Memory Acronym: "Match the surname to the state art."***Examiner's Trap:** Asks classical dance form of lesser-known exponents.

Speed Weapon: • Maharaj / Kalka-Bindadin → Kathak (UP)

- Reddy / Raja & Radha → Kuchipudi (Andhra)
- Mohapatra / Sapat / Pani → Odissi (Odisha)
- Jhaveri Sisters / Singh → Manipuri (Manipur)
- Sarma / Saikia / Borbayan → Sattriya (Assam)

Live Example & Execution: Question asks about Guru Bipin Singh → 'Singh' → **Manipuri**. Shovana Narayan → **Kathak**. (Solved in 1s)

G-4: Polity Constitutional Articles (+89/+90 Center-to-State Rule)

Speed Target: 1-3s

Memory Acronym: "Center Article + 89 = State Equivalent."

Examiner's Trap: Forgetting State executive/judiciary articles.

Speed Weapon: Add 89 (or 90 for higher courts) to the Union Article number to get the corresponding State Article!

Live Example & Execution: • Attorney General (Art 76) + 89 = **Advocate General (Art 165)**.

• President Ordinance (Art 123) → Rearrange: **Governor Ordinance (Art 213)**. (Solved in 2s)

G-5: The 'Always / Never' Extreme Word Filter

Speed Target: 2-3s

Memory Acronym: "Absolutes lie, qualifiers fly."

Examiner's Trap: Lengthy 4-statement Polity or Economics questions.

Speed Weapon: Statements containing 'Always', 'Never', 'Completely', 'Solely' are FALSE in 90% of cases. Statements with 'May', 'Can', 'Some' are TRUE in 90% of cases.

Live Example & Execution: Statement: 'The President can NEVER promulgate an ordinance when Parliament is in session.' → Contains 'NEVER' → Flagged FALSE in 2s!

6. THE 30+ TO 180+ SCORE TRANSFORMATION BLUEPRINT

To break out of the 30-mark rut and reach 180+, apply the **Sectional Time & Accuracy Protocol** in your daily practice:

Section	Current Reality (30-Mark Rut)	Target Speed Protocol (180+ Marks)	Target Score
Reasoning	Gets stuck in 1 hard series (4 mins lost). Draws circles for syllogisms.	Visual blitz first (Mirror/Dice/Notch in 6m) + 100-50 syllogism + 20s skip on series.	46 – 50 / 50
General Awareness	Spends 15 mins overthinking, changing right instincts to wrong guesses.	5-min pure recall sweep + 5-min elimination + 5-min quarantine hands off mouse.	36 – 42 / 50
English Comprehension	Reads every word of every sentence. Gets confused by voice tense changes.	Verb-only scan on Active/Passive + Noun-Pronoun PQRS mandatory pairs.	46 – 50 / 50
Quantitative Aptitude	Writes full school equations. Runs out of time after solving 8 questions.	11-divisibility on Mensuration + Value putting on Algebra/Trig + AB Ladder on %.	46 – 50 / 50
TOTAL	~30 – 40 Marks (Frustrated)	Master Speed Weapons Applied	174 – 192 / 200